

Deliverable 8 — Transfer Study

Domain

Education / long-running learning assistants.

Analogous memory problem

The assistant must retain durable learner information across sessions: mastered concepts, misconceptions, explanation preferences, learning goals, recurring errors, and unfinished practice tasks.

Invariants that remain valid

Selective admission; persistent memory; learner-scoped retrieval; ranking; bounded context construction; confidence/provenance; lifecycle maintenance.

Assumptions that break

Semantic similarity alone is insufficient. Educational memory needs structured skill/mastery state and evidence. A generic confidence score cannot replace evidence-backed correctness.

Storage changes

Use vector retrieval plus structured metadata such as learner_id, subject, skill_id, proficiency estimate, evidence timestamp/source, curriculum level, confidence, and review status.

Retrieval changes

Combine semantic search with learner, subject, skill and curriculum constraints. Add recency and evidence-quality signals.

Ranking changes

Rank by pedagogical usefulness: relevance, evidence strength, recency, uncertainty and curriculum alignment. Conflicting recent evidence should outrank stale similar text.

Context changes

Build bounded learner state sections: strengths, misconceptions, goals, recent evidence and uncertainty. Clearly separate facts from hypotheses.

Security changes

Retain isolation, but add role-aware access for learner/teacher/administrator/service processes, audit logging, deletion and retention controls.

Evaluation changes

Evaluate retrieval relevance, learner-state correctness, stale-memory handling, privacy/isolation and pedagogical usefulness.

Architecture component to remove

Remove the generic 'conversation' memory category as the domain abstraction. Replace it with evidence-backed educational entities such as mastery evidence, misconception, goal, preference and learning task.

Transfer conclusion

The enduring principle is a controlled memory lifecycle—admit, persist with provenance, retrieve scoped candidates, rank with domain evidence, construct bounded context, and maintain continuously. ChromaDB and conversation-specific schemas are project choices, not invariants.